

7)a)	Rate of decay
b)i)	$N = \frac{1}{222} \times 6.02 \times 10^{23}$ $N = 2.71 \times 10^{21}$
ii)	$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{3.8 \times 24 \times 60 \times 60} = 2.11 \times 10^{-6}$
	$A = \lambda N = (2.11 \times 10^{-6})(2.71 \times 10^{21}) = 5.73 \times 10^{15}$
iii)	$A = A_0 e^{-\lambda t}$ $0.01 = e^{-\lambda t}$
	$t = - \frac{\ln(0.01)}{\ln 2}$ $= \frac{3.8 \times 24 \times 60 \times 60}{2.18} = 2.18 \times 10^6 \text{ s}$
iv)	$\Delta m = (222.017576 - 218.008966 - 4.002602)$ $= 0.006008u$
	$E = \Delta mc^2$ $= 0.006008(1.66 \times 10^{-27})(3 \times 10^8)^2 = 8.98 \times 10^{-13} \text{ J}$
	$E = N \Delta mc^2$ $= \frac{(2.71 \times 10^{21}) 8.98 \times 10^{-13}}{1.6 \times 10^{-19}} = 1.52 \times 10^{28} \text{ eV} = 1.52 \times 10^{22} \text{ MeV}$
v)	As the area of the GM tube cannot fully enclose the sample and therefore it can only capture a fraction of the total activity

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8(a)	$E = 28\,368\,2\,230\,8\,499$
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